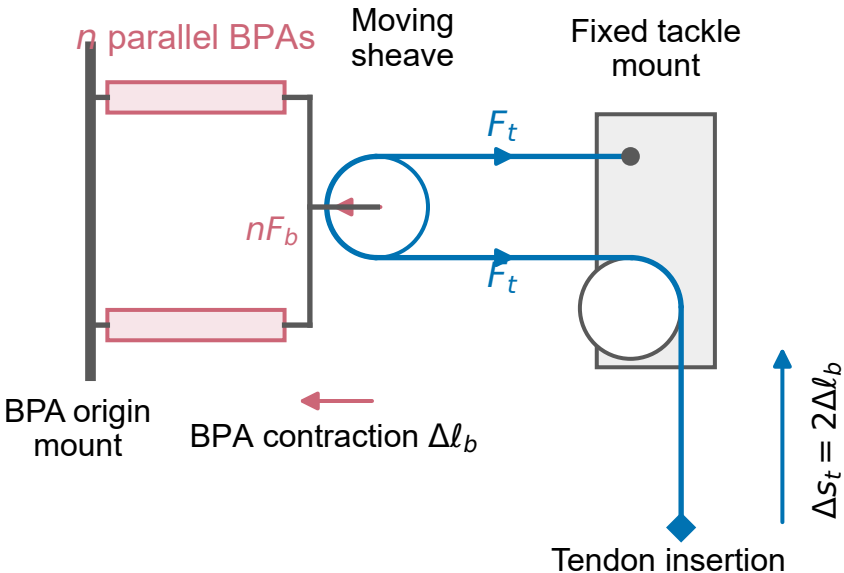
(A) Reverse tackle: two line parts



(B) Ideal relations

Travel multiplier

$$G = \Delta s_t / \Delta \ell_b$$

Force multiplier

$$F_t / F_b = n / G$$

Fixed tackle cable load

$$\mathbf{R}_c = nF_b\hat{\mathbf{u}}_b + F_t\hat{\mathbf{u}}_t$$

$\hat{\mathbf{u}}_b$: left; $\hat{\mathbf{u}}_t$: down

Support reaction: $-\mathbf{R}_c$

Separate BPA origin load

Ideal cable and rigid blocks shown. Compliance and pulley losses are omitted.